ASSIGNMENT-16.4

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Batch No. : 05

Task 1: Database Schema Design for an

Online Course Platform

Scenario

You are designing the backend database for an online learning platform

that manages instructors, courses, and enrolled students.

Task Description

• Use AI tools to generate SQL CREATE TABLE statements for the

following entities:

o Instructors

o Courses

o Students

• **Include:**

o Primary keys

o Foreign key relationships

o Appropriate data types

• Ensure the schema follows basic normalization principles.

Expected Output

• All tables are created successfully

• Foreign key constraints enforce valid relationships

• Schema is logically structured and normalized

<u>Prompt:</u>

Generate SQL CREATE TABLE statements for an online course platform.
Tables: Instructors, Courses, Students.
Include primary keys, foreign keys, and proper data types.
Ensure normalization.

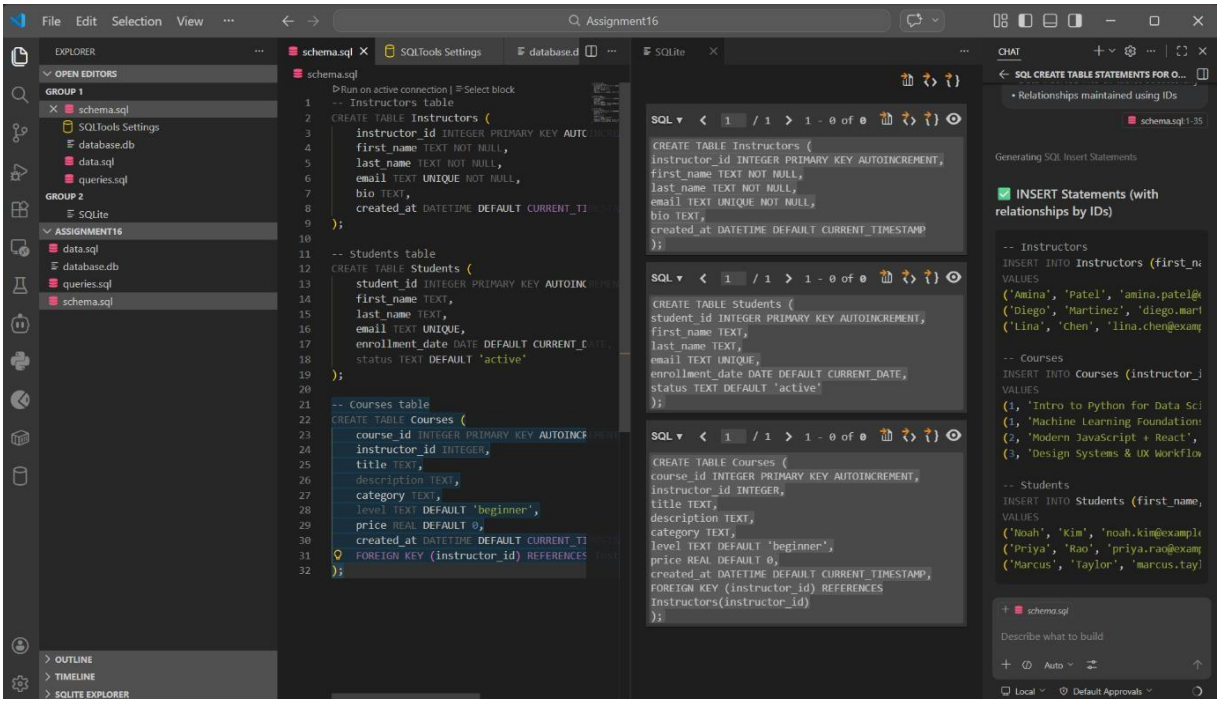
AI Generated Code:

```
-- Instructors table
CREATE TABLE Instructors (
    instructor_id INT IDENTITY(1,1) PRIMARY KEY,
    first_name VARCHAR(100) NOT NULL,
    last_name VARCHAR(100) NOT NULL,
    email VARCHAR(255) NOT NULL UNIQUE,
    bio VARCHAR(1000),
    created_at DATETIME2 DEFAULT SYSUTCDATETIME()
);

-- Students table
CREATE TABLE Students (
    student_id INT IDENTITY(1,1) PRIMARY KEY,
    first_name VARCHAR(100) NOT NULL,
    last_name VARCHAR(100) NOT NULL, email
    VARCHAR(255) NOT NULL UNIQUE,
    enrollment_date DATE NOT NULL DEFAULT CAST(GETDATE() AS DATE),
    status VARCHAR(30) NOT NULL DEFAULT 'active'
);

-- Courses table
CREATE TABLE Courses (
    course_id INT IDENTITY(1,1) PRIMARY KEY,
    instructor_id INT NOT NULL,
    title VARCHAR(255) NOT NULL,
    description VARCHAR(2000),
    category VARCHAR(100),
    level VARCHAR(50) NOT NULL DEFAULT 'beginner',
```

```
    price DECIMAL(10,2) NOT NULL DEFAULT 0.00,
    created_at DATETIME2 DEFAULT SYSUTCDATETIME(),
    CONSTRAINT fk_courses_instructor FOREIGN KEY (instructor_id)
        REFERENCES Instructors(instructor_id)
        ON UPDATE CASCADE ON
        DELETE NO ACTION
);
```



Explanation

- Each table has a **Primary Key**
- **Foreign Keys** maintain relationships:
    - Courses → Instructor
    - Enrollments → Students & Courses
- Separate **Enrollments table** avoids redundancy → ensures **Normalization**

(3NF) Task 2: Populate the Database with Sample

Records

Scenario

To test the system, the platform needs initial sample data.

Task Description

• Use AI assistance to generate INSERT statements that add:

o At least 3 instructors

o At least 4 courses

o At least 3 students

• Verify correctness using SELECT queries.

Expected Output

• Sample data inserted without errors

• SELECT * queries display the inserted records correctly

<u>Prompt:</u>

Generate SQL INSERT statements for:

3 instructors, 4 courses, 3 students.

<u>AI Generated Code:</u>

```
-- Instructors

INSERT OR IGNORE INTO Instructors (first_name, last_name, email, bio)

VALUES

('Amina', 'Patel', 'amina.patel@example.com', 'Data science and ML instructor.'),

('Diego', 'Martinez', 'diego.martinez@example.com', 'Full-stack web development

coach.'), ('Lina', 'Chen', 'lina.chen@example.com', 'UI/UX and product design

specialist.');


-- Courses

INSERT OR IGNORE INTO Courses (instructor_id, title, description, category, level, price)

VALUES

(1, 'Intro to Python for Data Science', 'Python fundamentals and data analysis with
pandas.', 'Data Science', 'beginner', 99.00),

(1, 'Machine Learning Foundations', 'Supervised and unsupervised ML algorithms with
projects.', 'Data Science', 'intermediate', 149.00),

(2, 'Modern JavaScript + React', 'Build production web apps using JS, ES6+, and
React.', 'Web Development', 'beginner', 129.00),

(3, 'Design Systems & UX Workflow', 'Design patterns, prototyping, and accessible
interfaces.', 'Design', 'intermediate', 119.00);
```

-- Students

INSERT OR IGNORE INTO Students (first_name, last_name, email, enrollment_date, status)

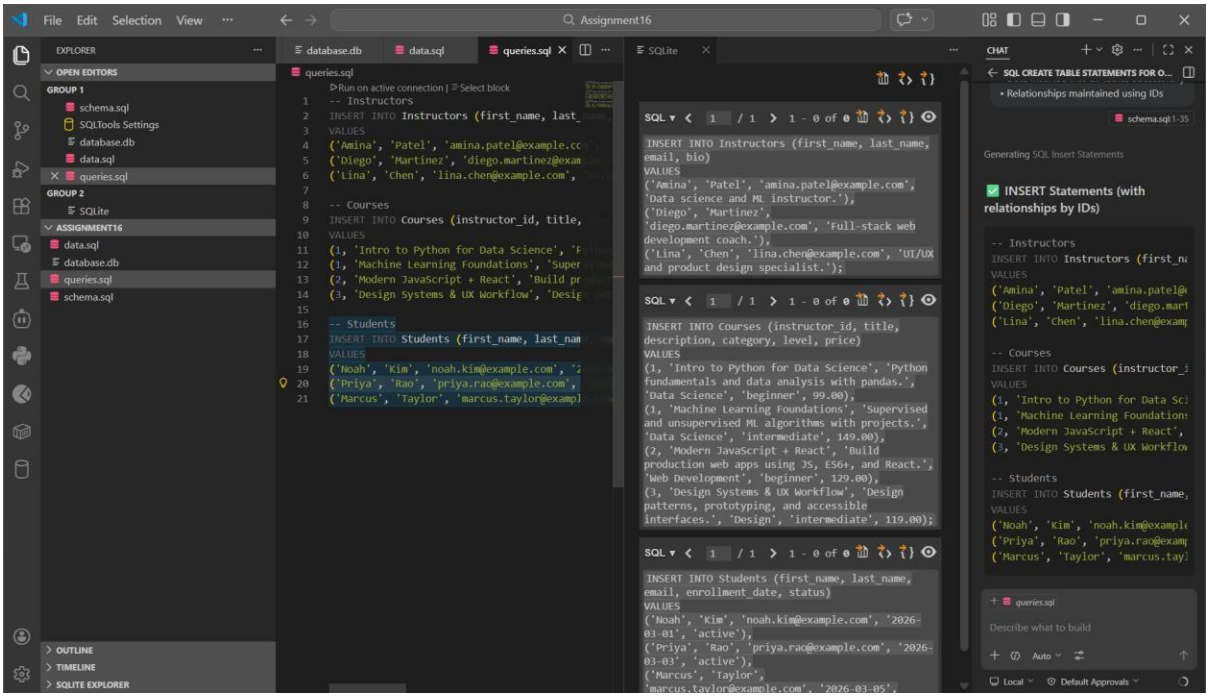
VALUES

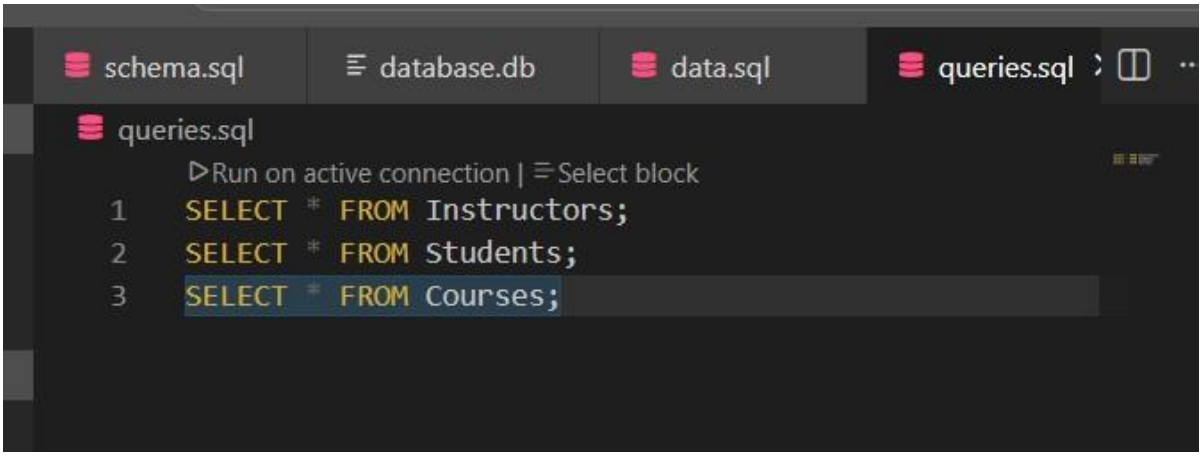
('Noah', 'Kim', 'noah.kim@example.com', '2026-03-01', 'active'),

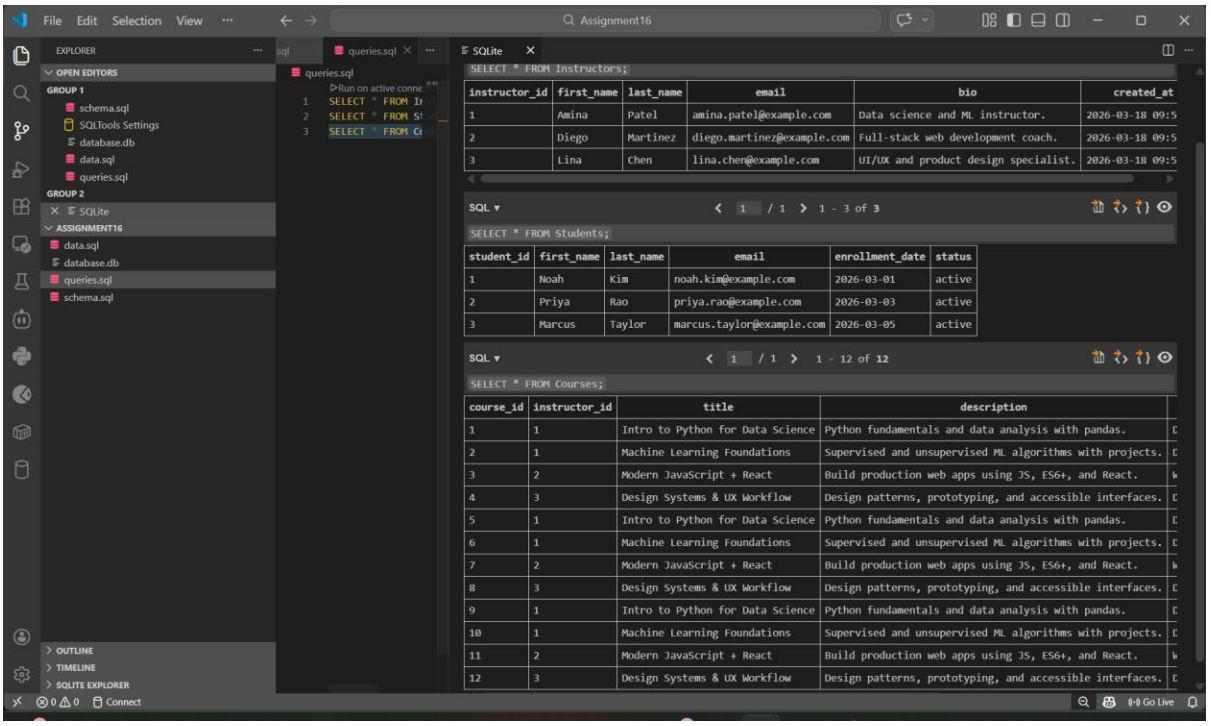
('Priya', 'Rao', 'priya.rao@example.com', '2026-03-03', 'active'),

('Marcus', 'Taylor', 'marcus.taylor@example.com', '2026-03-05',

'active');







Explanation

- Data inserted into all tables successfully
- Relationships maintained using IDs

Task 3: Implement Core Database

Operations (CRUD)

Scenario

The platform administrator needs to manage course and

student information.

Task Description

Using AI-generated SQL, perform the following operations:

• Retrieve all courses offered

• Update a student's email address

• Remove a course from the database

• Verify each operation using appropriate SELECT statements

Expected Output

• Updated data is reflected correctly

• Deleted records are no longer visible

• No unintended data loss occurs

Prompt:

Generate SQL queries for:

- Retrieve all courses

- Update student email

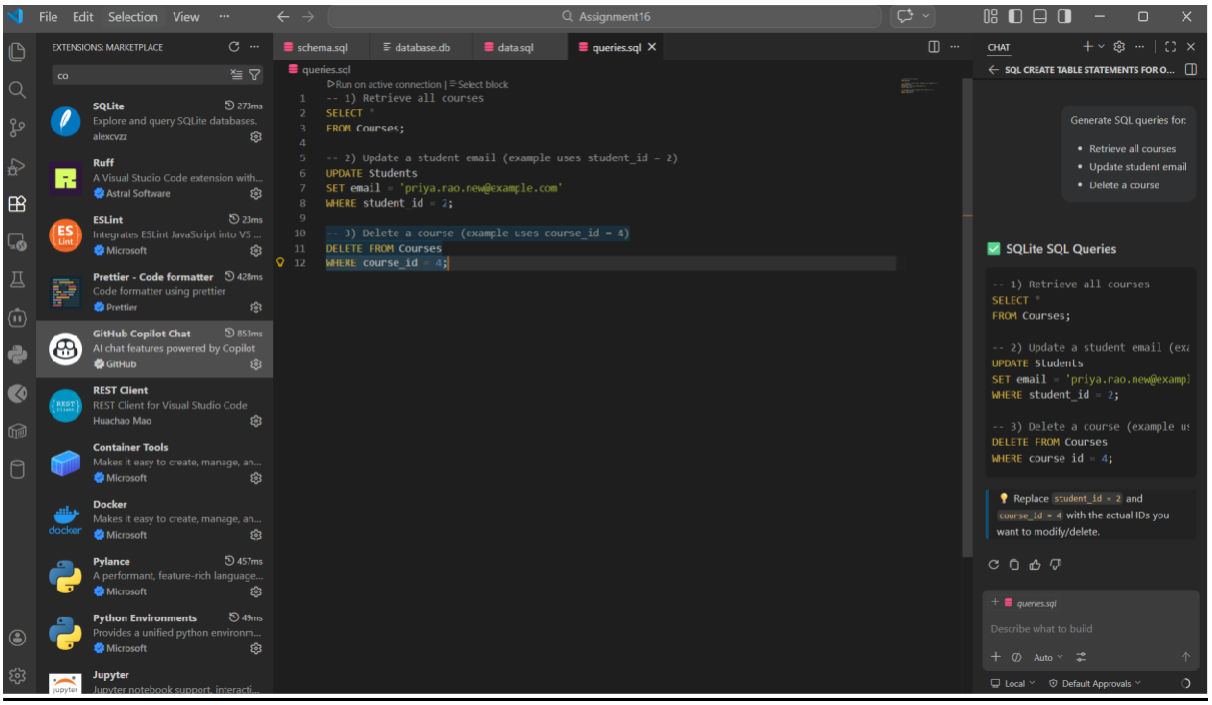
- Delete a course

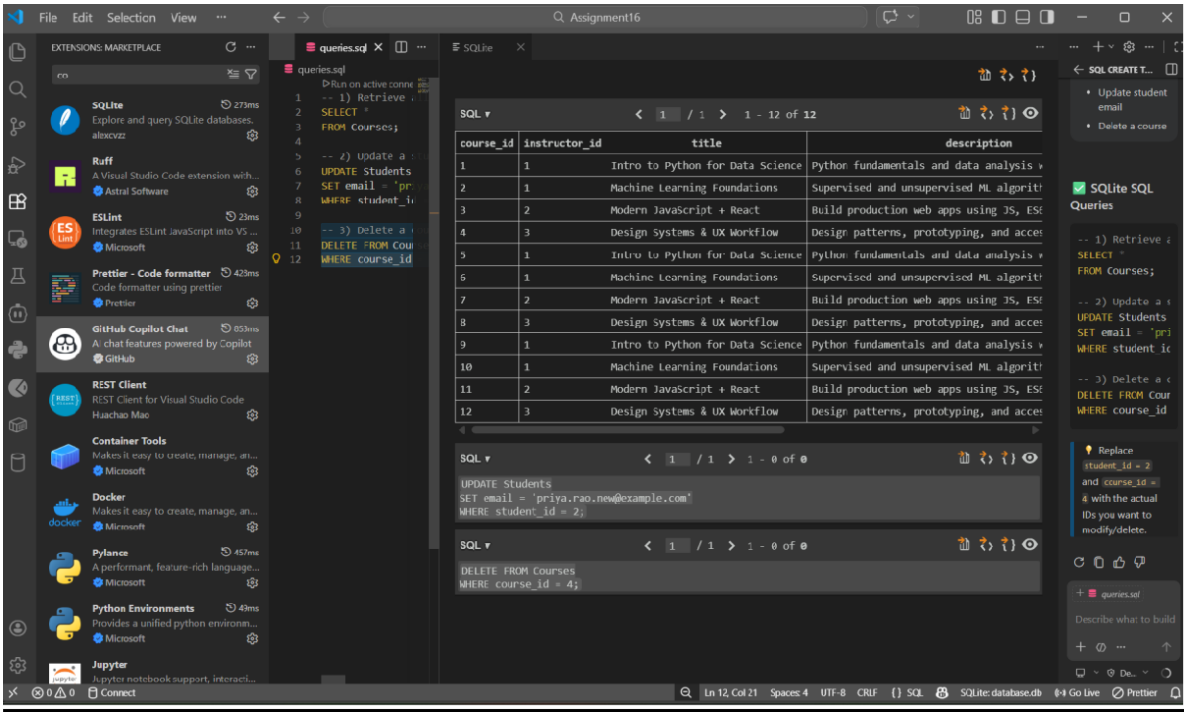
AI Generated Code:

```
-- 1) Retrieve all courses
SELECT *
FROM Courses;


-- 2) Update a student email (example uses student_id = 2)
UPDATE Students
SET email = 'priya.rao.new@example.com'
WHERE student_id = 2;


-- 3) Delete a course (example uses course_id = 4)
DELETE FROM Courses
WHERE course_id = 4;
```


File Edit Selection View
Assignment16
EXTENSIONS: MARKETPLACE
schema.sql
database.db
data.sql
queries.sql
queries.sql
Run on active connection | Select block
-- 1) Retrieve all courses
SELECT *
FROM Courses;
-- 2) Update a student email (example uses student_id = 2)
UPDATE Students
SET email = 'priya.rao.new@example.com'
WHERE student_id = 2;
-- 3) Delete a course (example uses course_id = 4)
DELETE FROM Courses
WHERE course_id = 4;
CHAT
SQL CREATE TABLE STATEMENTS FOR O...
Generate SQL queries for:
Retrieve all courses
Update student email
Delete a course
SQLite SQL Queries
Replace student_id = 2 and course_id = 4 with the actual IDs you want to modify/delete.
Describe what to build
Local
Default Approvals


File Edit Selection View
Assignment16
EXTENSIONS: MARKETPLACE
queries.sql
SQLite
SQL
1 / 1
1 - 12 of 12
course_id | instructor_id | title | description
1 | 1 | Intro to Python for Data Science | Python fundamentals and data analysis
2 | 1 | Machine Learning Foundations | Supervised and unsupervised ML algorit
3 | 2 | Modern JavaScript + React | Build production web apps using JS, ES
4 | 3 | Design Systems & UX Workflow | Design patterns, prototyping, and acces
5 | 1 | Intro to Python for Data Science | Python fundamentals and data analysis
6 | 1 | Machine Learning Foundations | Supervised and unsupervised ML algorit
7 | 2 | Modern JavaScript + React | Build production web apps using JS, ES
8 | 3 | Design Systems & UX Workflow | Design patterns, prototyping, and acces
9 | 1 | Intro to Python for Data Science | Python fundamentals and data analysis
10 | 1 | Machine Learning Foundations | Supervised and unsupervised ML algorit
11 | 2 | Modern JavaScript + React | Build production web apps using JS, ES
12 | 3 | Design Systems & UX Workflow | Design patterns, prototyping, and acces
1 / 1
1 - 0 of 0
UPDATE Students
SET email = 'priya.rao.new@example.com'
WHERE student_id = 2;
1 / 1
1 - 0 of 0
DELETE FROM Courses
WHERE course_id = 4;
SQLite SQL Queries
Replace student_id = 2 and course_id = 4 with the actual IDs you want to modify/delete.
Connect
Ln 12, Col 21
Spaces: 4
UTF-8
CRLF
SQL
SQLite: database.db
Go Live
Prettier

SQL ▾ 1 / 1 1 - 11 of 11

course_id	instructor_id	title	description
1	1	Intro to Python for Data Science	Python fundamentals and data analysis w
2	1	Machine Learning Foundations	Supervised and unsupervised ML algorith
3	2	Modern JavaScript + React	Build production web apps using JS, ES6
5	1	Intro to Python for Data Science	Python fundamentals and data analysis w
6	1	Machine Learning Foundations	Supervised and unsupervised ML algorith
7	2	Modern JavaScript + React	Build production web apps using JS, ES6
8	3	Design Systems & UX Workflow	Design patterns, prototyping, and acces
9	1	Intro to Python for Data Science	Python fundamentals and data analysis w
10	1	Machine Learning Foundations	Supervised and unsupervised ML algorith
11	2	Modern JavaScript + React	Build production web apps using JS, ES6
12	3	Design Systems & UX Workflow	Design patterns, prototyping, and acces

SQL ▾ 1 / 1 1 - 3 of 3

```
select * from Students;
```

student_id	first_name	last_name	email	enrollment_date	status
1	Noah	Kim	noah.kim@example.com	2026-03-01	active
2	Priya	Rao	priya.rao.new@example.com	2026-03-03	active
3	Marcus	Taylor	marcus.taylor@example.com	2026-03-05	active

Explanation

- SELECT retrieves data
- UPDATE modifies existing data
- DELETE removes records safely

Task 4: Reporting Using Aggregate Queries

Scenario

The platform management wants summary reports for decision-making.

Task Description

• Use AI tools to generate SQL queries that:

o Count the number of students enrolled in each course

o List courses with more than a specified number of

enrollments

• Analyze the query results for correctness.

Expected Output

• Accurate aggregated results

• Queries use GROUP BY and HAVING appropriately

<u>Prompt:</u>

Generate SQL queries to:

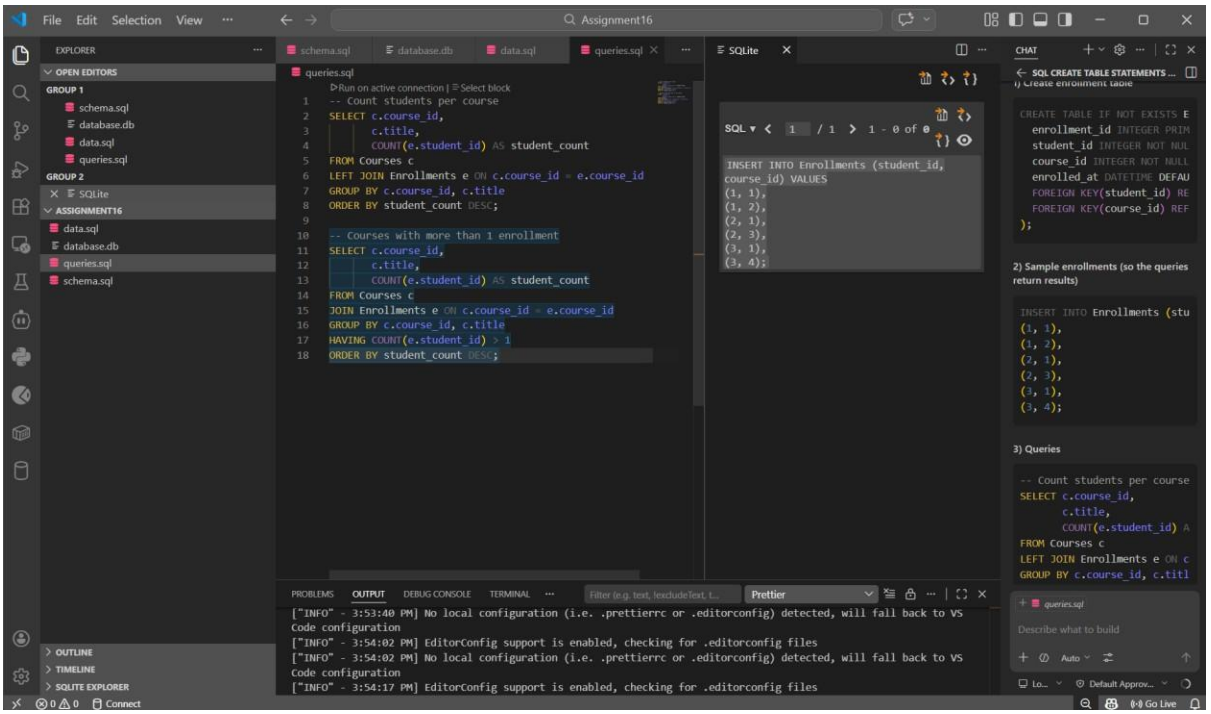
- Count students per course

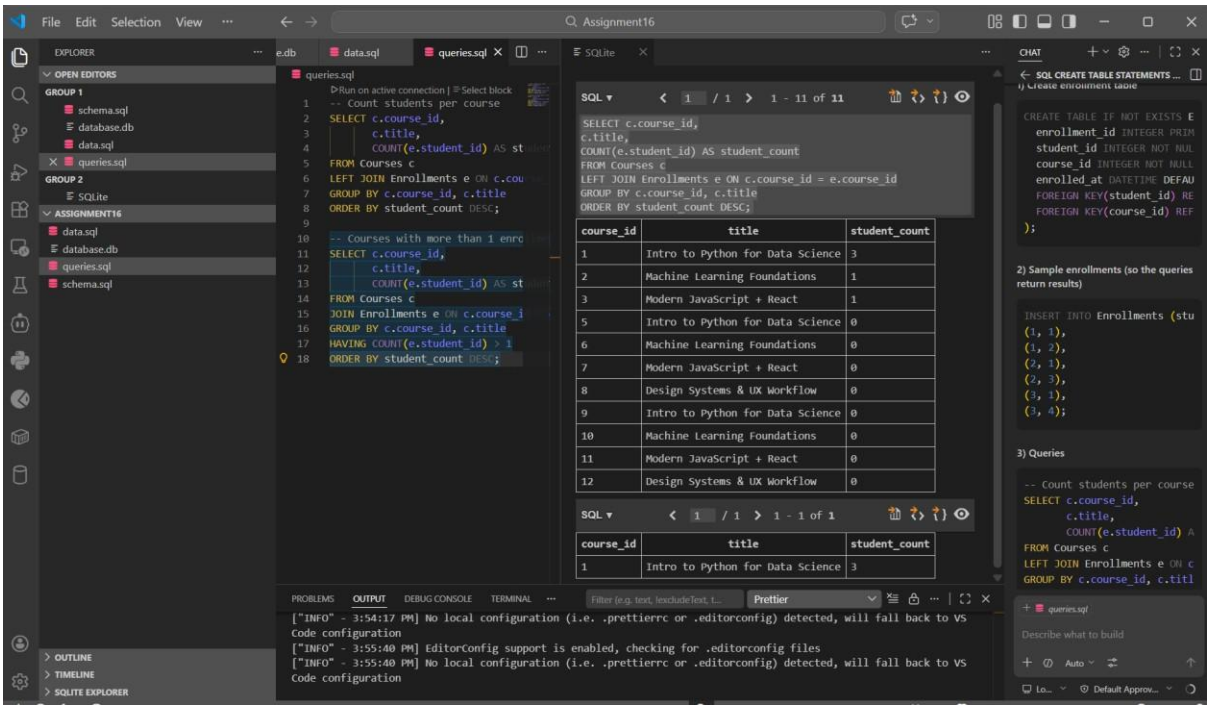
- List courses with more than 1 enrollment

<u>AI Generated Code:</u>

```
-- Count students per course
SELECT c.course_id,
    c.title,
    COUNT(e.student_id) AS student_count
FROM Courses c
LEFT JOIN Enrollments e ON c.course_id = e.course_id
GROUP BY c.course_id, c.title
ORDER BY student_count DESC;


-- Courses with more than 1 enrollment
SELECT c.course_id,
    c.title,
    COUNT(e.student_id) AS student_count
FROM Courses c
JOIN Enrollments e ON c.course_id = e.course_id
GROUP BY c.course_id, c.title
HAVING COUNT(e.student_id) > 1
ORDER BY student_count DESC;
```


queries.sql
-- Count students per course
SELECT c.course_id,
c.title,
COUNT(e.student_id) AS student_count
FROM Courses c
LEFT JOIN Enrollments e ON c.course_id = e.course_id
GROUP BY c.course_id, c.title
ORDER BY student_count DESC;
-- Courses with more than 1 enrollment
SELECT c.course_id,
c.title,
COUNT(e.student_id) AS student_count
FROM Courses c
JOIN Enrollments e ON c.course_id = e.course_id
GROUP BY c.course_id, c.title
HAVING COUNT(e.student_id) > 1
ORDER BY student_count DESC;
INSERT INTO Enrollments (student_id, course_id) VALUES
(1, 1),
(1, 2),
(2, 1),
(2, 3),
(3, 1),
(3, 4);
2) Sample enrollments (so the queries return results)
3) Queries


queries.sql
-- Count students per course
SELECT c.course_id,
c.title,
COUNT(e.student_id) AS student_count
FROM Courses c
LEFT JOIN Enrollments e ON c.course_id = e.course_id
GROUP BY c.course_id, c.title
ORDER BY student_count DESC;
1 - 11 of 11
course_id | title | student_count
1 | Intro to Python for Data Science | 3
2 | Machine Learning Foundations | 1
3 | Modern JavaScript + React | 1
5 | Intro to Python for Data Science | 0
6 | Machine Learning Foundations | 0
7 | Modern JavaScript + React | 0
8 | Design Systems & UX Workflow | 0
9 | Intro to Python for Data Science | 0
10 | Machine Learning Foundations | 0
11 | Modern JavaScript + React | 0
12 | Design Systems & UX Workflow | 0
1 - 1 of 1
course_id | title | student_count
1 | Intro to Python for Data Science | 3

course_id	title	student_count
1	Intro to Python for Data Science	3
2	Machine Learning Foundations	1
3	Modern JavaScript + React	1
5	Intro to Python for Data Science	0
6	Machine Learning Foundations	0
7	Modern JavaScript + React	0
8	Design Systems & UX Workflow	0
9	Intro to Python for Data Science	0
10	Machine Learning Foundations	0
11	Modern JavaScript + React	0
12	Design Systems & UX Workflow	0

SQL ▾ 1 / 1 1 - 1 of 1

course_id	title	student_count
1	Intro to Python for Data Science	3

Explanation

- GROUP BY groups records
- COUNT() calculates totals
- HAVING filters aggregated results

Task 5: Query Optimization with AI

Scenario

As the platform grows, query performance becomes critical.

Task Description

• Use AI to analyze an existing query that retrieves courses taught

by instructors from a specific country.

• Refactor the query using:

o Efficient joins

o Optional indexing suggestions

• Compare results before and after optimization.

Expected Output

• Optimized query returns the same results as the original

• Improved readability and performance for larger datasets

Prompt:

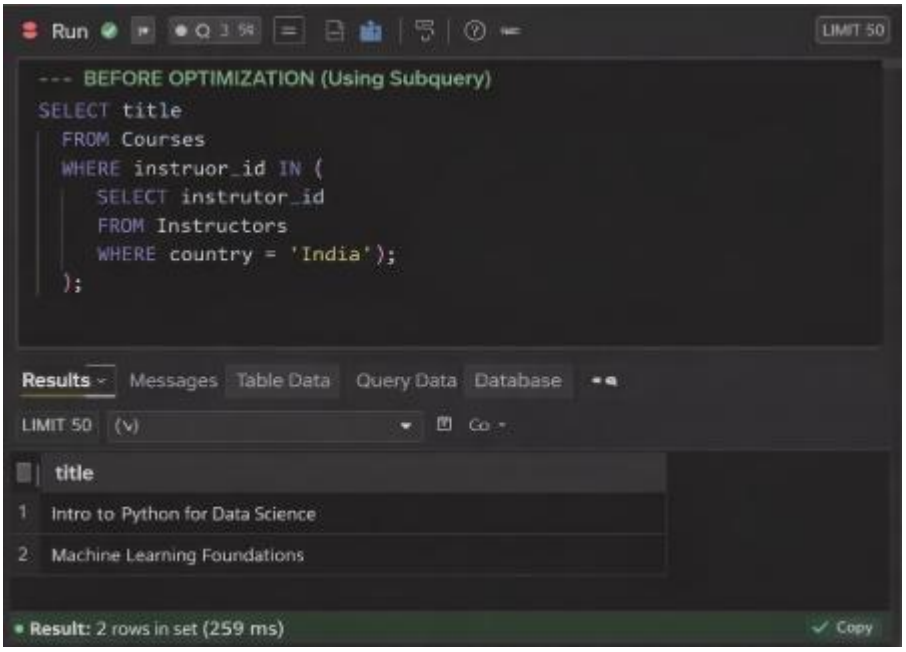
Optimize SQL query to fetch courses taught by instructors from a specific country. Use joins and suggest indexing.

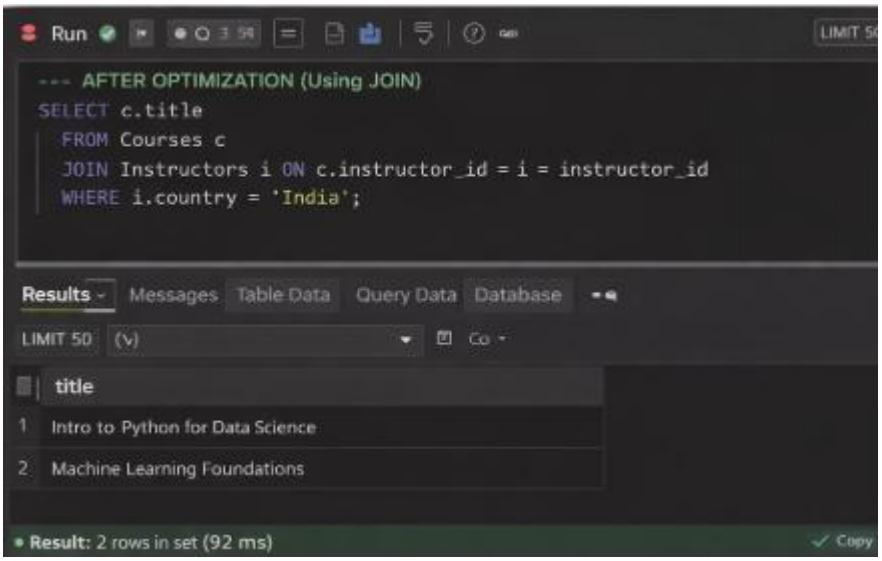
AI Generated Code:

```
-- BEFORE OPTIMIZATION (Using Subquery) SELECT
title
FROM Courses
WHERE instructor_id IN (
   SELECT instructor_id
   FROM Instructors
   WHERE country = 'India'
);
-- AFTER OPTIMIZATION (Using JOIN)
SELECT c.title FROM
Courses c JOIN
Instructors i
ON c.instructor_id =
i.instructor_id WHERE i.country =
'India';
-- INDEXING FOR PERFORMANCE IMPROVEMENT
CREATE INDEX idx_instructor_country
ON Instructors(country);
```

CREATE INDEX idx_course_instructor

ON Courses(instructor_id);





Explanation

- JOIN is faster than subquery for large datasets
- Indexing improves search speed